

Simplicial and singular homology groups

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1 Understanding the shape of data

When analysing data, being able to understand its general "shape" is massively helpful and important, as it enables us to potentially reduce the dimensionality of the dataset or craft more poignant features with respect to some classification task that is of interest.

We've already seen that a lot of the information of shape is captured quite well by the topological space that we assume our data to lie in. Furthermore there exists a host of mathematical tools for the analysis and differentiation of topological spaces that have a meaningful and relevant interpretation in this context of shape. Perhaps the most important such tools with regards to topological data analysis, due to both their relatively intuitive meaning and ease of computation, are the (singular/simplicial) homology groups of a space.

Roughly speaking these groups capture the number of "n-dimensional holes" (the rank of the nth homology group or Betti-number) that a space has plus some kind of torsion which, while a little harder to interpret, might also provide insight into said space's structure.

2 Simplicial homology

We have already learned about the concept of simplicial complexes and how we can use them to build a large variety of topological spaces from very simple building blocks (fittingly called simplices). In particular we have seen that the topology of a (geometric) simplicial complex is (up to homeomorphism) uniquely determined by a purely combinatorial description, the abstract simplicial complex, and that for every structure preserving (i.e. simplicial) map between these abstract complexes we can construct a corresponding continuous map between any of their geometric realizations that maps its vertices in the same way the simplicial map does. Conversely and even more profoundly to any continuous map ϕ we can also assign a simplicial map ϕ' that in a very precise sense approximates ϕ (its aforementioned geometric realization $|\phi'|$ is homotopic to ϕ).

All this serves as a justification for why, in attempting to understand the topology, we shall restrict ourselves to analysing the graph-like structure of an abstract simplicial complex (or respectively of the slightly more general Δ -complex), using simplicial homology. While there is still some ambiguity in the fact that we can find different simplicial complex structures for one and the same space (think subdivision), ultimately this connection can be further corroborated by introducing the so called singular homology groups which can be defined for arbitrary topological spaces and for whom it is possible to show that they coincide with their simplicial counterparts if the underlying space can indeed be described by an abstract simplicial complex. But that proof requires

a great deal of additional mathematical machinery, which would extend far beyond the scope of this paper (for additional reading in that direction consider Nadathur[2]).

2.1 Oriented simplices

Key in defining simplicial homology is the idea of giving simplices an orientation. In the simple case of a 1-simplex $\{v_1, v_2\}$ (imagine a line segment) this amounts to thinking of the line being an arrow that can point in either of two directions, each of which corresponds to one of the two orderings of its vertices (v_1, v_2) and (v_2, v_1) . For general n -simplices there can be more than two orderings of its vertices, but for our purposes we would only like to differentiate between two orientations.

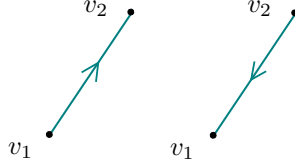


Figure 1 – Two different orientations of a 1-simplex, corresponding to its two orderings

This is achieved by defining two orderings of an n -simplex Δ^n to be equivalent iff they differ by an even permutation $\tau \in S_{\Delta^n}$, $\text{sgn}(\tau) = 1$. To gain an intuition for what this means consider a 2-simplex $\{v_1, v_2, v_3\}$ (imagine a triangle). Given an ordering, say (v_1, v_2, v_3) , we can picture tracing along the (oriented) 1-simplices that make up its boundary, going first from v_1 to v_2 via (v_1, v_2) , then from v_2 to v_3 via (v_2, v_3) and finally from v_3 to v_1 . If another ordering arises from simply swapping two of the vertices, i.e. a transposition, say (v_1, v_3, v_2) then doing the same thing with the new order amounts to going around the simplex in the opposite direction. Applying another transposition, will result in an ordering that now differs from the first one by an even permutation and doing the same procedure as before also moves us around the simplex in the same direction again. Thus we can think of the two kinds of orderings of a 2-simplex as representing clockwise and counterclockwise orientation. It's worth noting that we don't need to worry about which is which exactly. We are only interested in when two orderings are different.

Knowing what an oriented simplex is, we shall from now on consider only oriented simplices with some arbitrary orientation chosen at their inception.

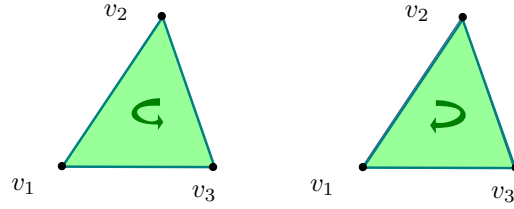


Figure 2 – The orientations of a 2-simplex; possible corresponding orderings could be (v_1, v_2, v_3) and (v_1, v_3, v_2)

2.2 The group of k -chains

To understand how oriented simplices are helpful, we will first have an informal look at an idea that, if just formally introduced, will seem rather arcane and abstract.

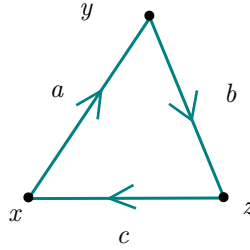


Figure 3 – A simplicial complex showing an empty triangle

What we are going to want to be able to do, is to "add" (abstract) simplices of the same dimension together. Doing so allows us to, in a sense, talk about this "walking along" a sequence of k -simplices, as already alluded to in the previous section. In figure 3 for instance if we say $a + b + c$, we can think about first going along a , then b , then c , forming a kind of 1-dimensional *chain*. This chain in particular actually forms a loop. And what do loops go around? Holes (the plot thickens)!

So the next step is to consider how exactly we go about detecting loops. The idea is to look at the boundary of a chain. Let's recall that the boundary of a single simplex is the set of its faces. Since our new addition will separately be defined on the sets of k -simplices for each k , we could also now say that the boundary of a simplex is the *sum* of its faces (e.g. $\partial a = x + y$). And we might further define the boundary of a chain (i.e. a sum of simplices) to be the sum of the boundaries of each of the simplices. That would mean that for instance $\partial(a + b) = \partial a + \partial b = (x + y) + (y + z) = x + 2y + z$ (the last step assuming

associativity). Now that doesn't seem quite right yet. Intuitively it would make much more sense if the boundary of $a+b$ would only involve x and z . And this is where, finally, orientation comes into play. The boundary of a simplex won't just be the sum of its faces but we will give each of the summands either an additional minus or plus sign, depending on the original simplex's orientation. For a 1-simplex such as a that means assigning a plus to the "start"- and a minus to the "end"-point. So $\partial a = x - y$ and (!) $\partial(a+b) = \partial a + \partial b = (x - y) + (y - z) = x - z$. Bingo bongo! And now what does the boundary of the (cyclic) chain $a + b + c$ turn out to be?

$$\partial(a + b + c) = \partial a + \partial b + \partial c = (x - y) + (y - z) + (z - x) = 0$$

It's zero! All the start- and endpoints cancel each other out. And it should be fairly intuitive that this will hold for any cycle.

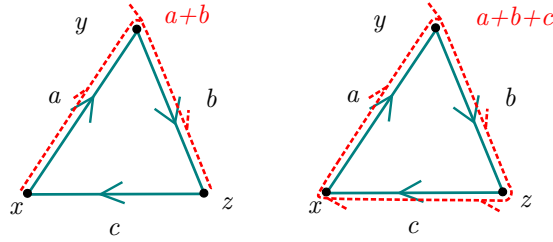


Figure 4 – A depiction of the chains $a + b$ and $a + b + c$, which forms a cycle and hence doesn't have a boundary

We are now actually really close already to the idea of a homology group. One last thing to note before defining all these concepts rigorously is that we want this addition to be commutative. This becomes pretty clear when we consider that we don't want to distinguish between different holes if their enclosing cycles start at a different point. Obviously $a + b + c$ and $b + c + a$ enclose the same hole. Thus we will, in this context, view them as the same chain. This also means that we can ignore potentially paradoxical sums like $a + c + b$ (you can't really first go along a , then c , since a doesn't end where c starts), as they will be identified with the likes of $a + b + c$.

2.2.1 Algebra interjection I: Free abelian groups

We recall that a group is a set G equipped with a binary operation $\circ : G \times G \rightarrow G$ that is associative and gives rise to a neutral group element e which in turn assigns every $g \in G$ an inverse $g^{-1} \in G$, such that $e \circ g = g \circ e = g$ and $g \circ g^{-1} = g^{-1} \circ g = e$. If additionally the group elements commute under $\circ$ the group is called abelian. In the case of abelian groups it is common to denote the

operation $+$ and indicate inverses with a minus sign, in analogy to the integers under addition, who also form an abelian group.

Looking over this definition we might recognize that we talked about sums of things, about a neutral element (0), about inverses (simplices of opposite orientation), about commutativity. So it already seems like abelian groups aren't suited too badly to what we set out to make precise. Really the last question is: What set are we operating on exactly? It turns out the key object of interest that will include the final piece of the puzzle is something called a *free abelian group*:

The free abelian group $F(M)$ generated by a set M is the set of functions from M into $\mathbb{Z}$ that are 0 almost everywhere and the group operation is pointwise addition of functions:

$$F(M) = \{f : M \rightarrow \mathbb{Z} \mid f(x) \neq 0 \text{ only for finitely many } x \in M\}$$

Basically we assign every element in M an integer. In that sense we can also think of elements of $F(M)$ as formal sums (this is more like what we've seen in the last section; a question to ask ourselves here would be what set M should be exactly, such that $F(M)$ will consist of all the k -chains), where almost all coefficients are 0:

$$\sum_{m \in M} a_m m, \quad a_m \in \mathbb{Z}$$

Or if M is finite as $|M|$ -tuples:

$$(a_1, a_2, \dots, a_{|M|}), \quad a_i \in \mathbb{Z}$$

The last view will hopefully evoke the image of an $\mathbb{R}$ -vector space where we only look at those vectors which have integer components and the dimension is the size of M , since we have exactly $|M|$ components in each vector. Connecting this view with that of formal sums, we can consider M to kind of be the basis of said vector space, which is why M is indeed also called the basis of $F(M)$. When building the intuition for what we are ultimately trying to accomplish we already saw that the view of formal sums emerges quite naturally in our context, so we will continue to use that notation. Anyways, now finally, to pull everything together, the long awaited definition of k -chains:

Suppose we are given an abstract simplicial complex S . Then for all $k \in \mathbb{Z}$ we define the group of k -chains C_k to be the free abelian group generated by the set of k -simplices in S where we further identify any simplex with the negation of the same simplex with opposite orientation, or more generally for arbitrary orderings of a simplex:

$$\Delta^k = \text{sgn}(\tau) \tau(\Delta^k)$$

where Δ^k is a k -simplex in S and τ is some permutation of Δ^k , applied element wise:

$$C_k = F(\{\Delta^k \text{ a } k\text{-simplex} \in S\})$$

2.3 Boundaries, cycles and the homology groups

All that's missing now is to extend the boundary operator beyond basic simplices to arbitrary k -chains in the way we already discussed in section 2.2:

$$\partial_k : C_k \rightarrow C_{k-1}, \quad \Delta^k \mapsto \sum_{i=0}^k (-1)^i (v_0, \dots, \hat{v}_i, \dots, v_k)$$

where Δ^k a k -simplex and $(v_0, \dots, \hat{v}_i, \dots, v_k)$ its face with the i th vertex removed. In other words it assigns to every k -simplex the formal sum of its faces in a particular orientation. Since the k -simplices form a basis of C_k this map extends naturally to a group homomorphism from C_k to C_{k-1} in a "linear" (to continue the analogy to vector spaces) way (i.e. if $\sigma = a_1 \Delta_1^k + a_2 \Delta_2^k + \dots + a_n \Delta_n^k \in C_k$ then $\partial_k \sigma = a_1 \partial_k \Delta_1^k + a_2 \partial_k \Delta_2^k + \dots + a_n \partial_k \Delta_n^k$, where Δ_i^k k -simplices in S).

Crucially, once again, if the boundary of a k -chain is 0, it means it loops back on itself, enclosing a k -dimensional hole in the k -skeleton of S . So these k -chains are the *cycles*: $Z_k := \ker(\partial_k)$ (we recall that the kernel is the set of elements in the domain that are mapped to zero). Now we're almost there! We already saw that Z_k effectively captures the nature of holes in the k -skeleton. But what if that hole is filled by a $(k+1)$ -simplex? For that purpose we will consider

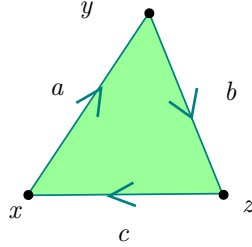


Figure 5 – The same triangle as in figure 3, filled out by a 2-simplex, making the chain $a + b + c$ homologous to 0

the boundaries of said $(k+1)$ -simplices or in other words the image of ∂_{k+1} : $B_k := \text{im}(\partial_{k+1})$. First off it's fairly obvious that *any* boundary in B_k is also a cycle in Z_k since it clearly encloses said simplex, i.e. it loops back on itself. We want to now set those cycles that arise from being a boundary of a higher simplex to zero, since they clearly don't actually represent a hole in the entire complex. We also say they are *homologous to 0*. Moreover in the group context this also means that we want all k -chains that only differ by such a boundary to be identified. How is this "setting to zero" achieved algebraically?

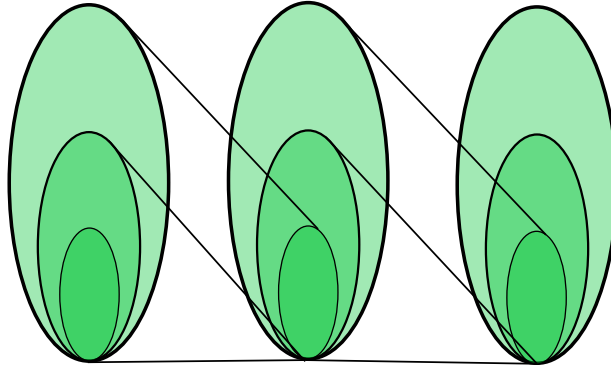


Figure 6 – A way to picture the different sets we are interested in: The outermost ovals represent the chain-groups for subsequent k 's. The nested ovals are the respective kernels of the boundary-operators, as they are sent to zero. And the innermost ovals are the images of the respective previous boundary-operator. They do in particular lie completely inside the kernel of the next one (obviously this picture can be extended arbitrarily far to the left and right).

2.3.1 Algebra interjection II: Normal subgroups and factor groups

Let G be a group and N be a subset of G , such that the group operation is close on N , i.e. that for any $n, n' \in N$, $nn' \in N$. Then we call N a *subgroup* of G . If furthermore holds that $gng^{-1} \in N$ for all $g \in G$ and $n \in N$, then we call N a *normal* subgroup of G . This condition is also called *conjugacy invariance* of N under G . Notice how this is always true if G is abelian since $gng^{-1} = gg^{-1}n = n \in N$.

If we have such a normal subgroup N then the *factor group* G/N is the set of equivalence classes of G such that elements of a class only differ by an $n \in N$. The condition on N to be normal allows us to naturally extend the group operation of G to G/N : To combine two equivalence classes in G/N simply take out representatives of G , combine those and find the equivalence class of the result.

Time for a very simple example: Take the integers $\mathbb{Z}$. We know that any of its integer multiples $n\mathbb{Z}$ are normal (since $\mathbb{Z}$ is abelian) subgroups. So what does, say $\mathbb{Z}/3\mathbb{Z}$, look like?

$$\mathbb{Z}/3\mathbb{Z} = \left\{ \begin{array}{l} \{\dots, -3, 0, 3, 6, \dots\}, \\ \{\dots, -2, 1, 4, 7, \dots\}, \\ \{\dots, -1, 2, 5, 8, \dots\} \end{array} \right\} = \left\{ \begin{array}{l} 3\mathbb{Z}, \\ 3\mathbb{Z} + 1, \\ 3\mathbb{Z} + 2 \end{array} \right\}$$

Everything that was originally in $3\mathbb{Z}$ now makes up the 0-element of the factor group and the other factor group elements are all cosets of $3\mathbb{Z}$, i.e. they arise from adding some $g \in \mathbb{Z}$ to all its members. If we restrict ourselves to a particular system of representatives, specifically $\{0, 1, 2\}$, then it becomes clear very

quickly that this group is precisely the *addition modulo 3*. To add two elements we use the original addition: $2 + 2 = 4$. Then we find that the representative that lies in the same coset as 4 is 1. Thus $2 + 2 = 1$ in the factor group, which would more commonly be denoted $2 + 2 \equiv 1 \pmod{3}$ to indicate what $n\mathbb{Z}$ exactly we modded out. The terminology of "modding out" a normal subgroup is thus actually used for arbitrary factor groups: G/N is often times referred to as " G modulo N " or even just " $G \pmod{N}$ ".

Finally, pulling everything together, all we need to show to apply this concept of modding out in our specific context is that

- (i) Z_k is itself a group
- (ii) B_k is a normal subgroup of Z_k

And these statements are corollaries of very basic group theory: The kernels and images of group homomorphisms $\phi : G \rightarrow H$ are subgroups of G and H respectively and any subgroup N' that is contained entirely within another subgroup N is also a subgroup of N . Normality of B_k following from abelianity and having already shown that $B_k = \text{im}(\partial_{k+1})$ does indeed lie in $Z_k = \ker(\partial_k)$ we can now define the simplicial k -th homology group of S :

$$H_k(S) = Z_k/B_k$$

2.4 Example calculation and interpretation of H_k

Enough defining! Let's actually compute the homology groups of a simplicial complex. We'll investigate the following 2-dimensional example complex:

$$S = \left\{ \begin{array}{l} (x_1), (x_2), (x_3), (x_4), (x_5), \\ (x_1, x_2), (x_2, x_3), (x_3, x_1), (x_1, x_4), (x_2, x_4), \\ (x_1, x_2, x_3) \end{array} \right\}$$

To make things easier to talk about we'll label the 1-simplices a, b, c, d and e .

So what are the relevant homology groups? Drawing out all the chain groups and their connecting boundary operators, we get the following picture (this structure is also called a *chain complex*; also have a look back at figure 6 to remember where the cycles- and boundary-groups are in there):

$$\dots \xrightarrow{\partial_5} C_4 \xrightarrow{\partial_4} C_3 \xrightarrow{\partial_3} C_2 \xrightarrow{\partial_2} C_1 \xrightarrow{\partial_1} C_0 \xrightarrow{\partial_0} C_{-1} \xrightarrow{\partial_{-1}} \dots$$

Now S doesn't contain any simplices of dimension > 2 and no complex will ever contain (since there aren't any) a simplex of negative dimension. So those groups are all trivial (the question of why a free abelian group over an empty set is trivial shall be a small exercise for the reader). The homology groups resulting from these chains are thus also all trivial and don't really carry any

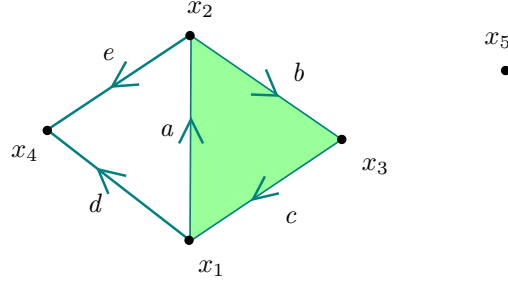


Figure 7 – A visualization of S

information. This allows us to simplify the picture, yielding the following chain complex:

$$0 \xrightarrow{\partial_3} C_2 \xrightarrow{\partial_2} C_1 \xrightarrow{\partial_1} C_0 \xrightarrow{\partial_0} 0$$

We can now see that what makes sense intuitively (i.e. that in a complex of some dimension n there can only be holes up to that dimension), is indeed true and the remaining homology groups to be computed are $H_2(S)$, $H_1(S)$ and $H_0(S)$.

We'll begin with H_0 : The kernel of ∂_0 is obviously the entire group, given that everything gets sent to zero and the image of ∂_{k+1} consists of the boundaries of all the 1-chains (i.e. all the linear combinations of 1-simplices):

$$B_0 = \{ \partial_0(\alpha a + \beta b + \gamma c + \delta d + \varepsilon e) \in C_0 \mid \alpha, \beta, \gamma, \delta, \varepsilon \in \mathbb{Z} \}$$

Using the definition of ∂_0 we can write any element of B_k with the following expression:

$$\begin{aligned} & \alpha \partial_0 a + \beta \partial_0 b + \gamma \partial_0 c + \delta \partial_0 d + \varepsilon \partial_0 e \\ &= \alpha(x_1 - x_2) + \beta(x_2 - x_3) + \gamma(x_3 - x_1) + \delta(x_1 - x_4) + \varepsilon(x_2 - x_4) \end{aligned}$$

Technically the 0-simplices are $(x_1), (x_2), \dots$, but I'll write them as just $x_1, x_2, \dots$ to simplify things. We don't ever really need to make a differentiation between a 0-simplex and the point it's made up of.

With the generating bases of C_1 and C_0 we see that we can therefore also represent ∂_1 as an integer matrix:

$$A = \begin{pmatrix} 1 & 0 & -1 & 1 & 0 \\ -1 & 1 & 0 & 0 & 1 \\ 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & 0 & -1 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

So $H_0(S) = Z_0/B_0 = C_0/AC_1 \cong \mathbb{Z}^5/A\mathbb{Z}^5$. We can show that this is in fact isomorphic to $\mathbb{Z}^2 = \mathbb{Z} \times \mathbb{Z}$, representing the fact that S has two connected

components (yes this is the meaning of 0-dimensional hole). We won't go into the details of that procedure, since it is quite mechanical and requires yet more algebra to understand. To provide an intuition though, it can help once again to think in analogy to vector spaces. If the image of an $\mathbb{R}$ -matrix is d -dimensional then it is isomorphic to $\mathbb{R}^d$. Similarly the "dimension" (technically called the *rank*) of an abelian group is basically the number of times you can factor a copy of $\mathbb{Z}$ out of it. And finding the rank of the image of an abelian group homomorphism works just like finding the dimension of the image of a linear map: It's the rank of any matrix that represents the map. Thus row-reducing A we get the following matrix:

$$\begin{pmatrix} 1 & 0 & -1 & 1 & 0 \\ 0 & 1 & -1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

And we see that it has rank 3. So if somehow B_0 consists of 5 copies of $\mathbb{Z}$ and Z_0 of 3, then it stands to reason that factoring one out of the other will indeed yield $\mathbb{Z} \times \mathbb{Z}$.

In general this is not the whole story. The integers, as opposed to a field such as the reals, are not closed under multiplication (disregarding 0). So AC_1 could be embedded non-trivially in $\mathbb{Z}^5$ (just like $2\mathbb{Z}$ as its own group is isomorphic to $\mathbb{Z}$). The good thing is that that doesn't change the rank of the factor group (if we factor $2\mathbb{Z}$ out of $\mathbb{Z}$, we are left with $\mathbb{Z}/2\mathbb{Z}$ which, having only two elements, can't contain another copy of $\mathbb{Z}$), so it won't make a difference in the number of holes. But it does add extra finite components to it. This is referred to as torsion, which in our example actually is trivial. How we can interpret torsion is not something we'll go into here.

Moving along to the second homology group H_1 we are now interested in the kernel of ∂_1 . Once again we can use the matrix A : We now need to solve the system of equations $Ax = 0$, where x is an integer-"vector". Having already transformed A into row-echelon-form we can see that there are two degrees of freedom and it should be a familiar task from linear algebra to figure out that the kernel is spanned by the following elements:

$$\begin{pmatrix} 1 \\ 0 \\ 0 \\ -1 \\ 1 \end{pmatrix} = a - d + e, \quad \begin{pmatrix} 1 \\ 1 \\ 1 \\ 0 \\ 0 \end{pmatrix} = a + b + c$$

This is exactly what we'd expect as $a - d + e$ and $a + b + c$ are clearly 1-cycles in S and the loop going all the way around $b + c + d - e$ is a linear combination of the two. So $Z_1 \cong \mathbb{Z} \times \mathbb{Z}$. In other words there are two 1-dimensional holes in the 1-skeleton, one corresponding to each factor $\mathbb{Z}$.

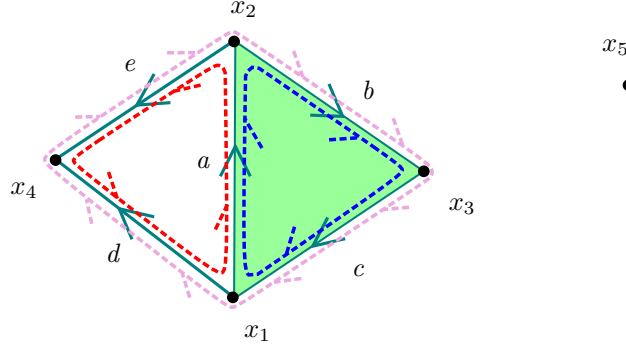


Figure 8 – The lilac cycle is a linear combinations of the blue and red cycles: $b + c + d - e = a + b + c - (a + e - d)$. Notice how choosing the proper orientation for the cycles results in a canceling out in the sum.

Now to B_1 : Since there is only one 2-simplex it's obvious that the image of ∂_2 is generated by that simplex's image: $\partial_2(x_1, x_2, x_3) = (x_2, x_3) - (x_1, x_3) + (x_1, x_2) = b - (-c) + a = a + b + c$. Setting this to zero in Z_1 leaves the cycle $a - d + e$ as the sole generator of $H_1(S)$, making it isomorphic to $\mathbb{Z}$. This is also exactly what we'd expect, given that the 2-simplex fills the hole surrounded by $a + b + c$ and S thus has exactly one 1-dimensional hole. Moreover this example also goes to show, how setting a cycle σ to zero makes other cycles whose difference is σ homologous: The two cycles $a + b + c$ and $b + c + d - e$ both go around the same hole, their difference is the null-homologous cycle $a - d + e$ and they are thus identified in $H_1(S)$.

Finally looking at the kernel of ∂_2 it is immediately obvious that it's zero, since the only 2-simplex is mapped to $a + b + c \neq 0$ (in the chain group). Therefore $H_2(S)$ is clearly also trivial, making its rank 0, which should again make sense since there are no 2-dimensional holes in S .

For a look at some more homology computations to get a better feel for the subject matter, especially with higher dimensional holes consider a video from N.J Wildberger on the subject[3].

2.5 Invariance of simplicial homology under homeomorphisms

As we discussed in the very beginning and as should also makes sense intuitively the number of holes and more generally the homology groups describe the topological space $|S|$. So continuously deforming $|S|$ into some other space via homeomorphism, that space should have the same simplicial homology. But to even talk about simplicial homology we first need a simplicial complex struc-

ture on the space again. Now we know that one exists, and we can also find one using triangulation, but it is not unique. Recall that we can for instance create arbitrarily large complexes out of S using subdivision, all of which give rise to homeomorphic topological spaces.

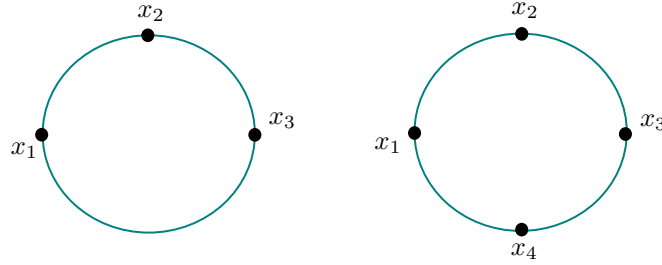


Figure 9 – On the left a circle with a simplicial complex structure; on the right we see that subdividing the complex yields a homeomorphic circle.

This means that if we want to be sure that the simplicial homology groups really do only give information about the topology and not about the - in our case mostly arbitrarily chosen - triangulation, then we have to show that complexes S and S' whose geometric realizations are homeomorphic also have the same homology groups. A sketch of that proof looks as follows:

1. If two complex's geometric realization $|S|$ and $|S'|$ are homeomorphic then there exists a bijective bicontinuous function $\phi : |S| \rightarrow |S'|$.
2. We can find a simplicial approximation $p : S \rightarrow S'$ of ϕ .
3. This approximation implies that S and S' have a common subdivision.
4. Show that subdividing a complex leaves the homology groups unchanged.

Alternatively there is another way to show the invariance of simplicial homology under homeomorphism, which is usually preferred because it also generalizes the idea of homology groups to arbitrary spaces, not just those that admit a simplicial complex structure. The disadvantage to this approach however, is that it requires a great deal of extra mathematical machinery and is pretty abstract. It involves first defining the so called singular homology groups of a space and then showing that if the space can be described by a simplicial complex that these coincide with the singular ones. The former step we are going to quickly have a look over for generality's sake since most papers discussing homology will refer to the singular one instead of the simplicial one and for the latter I will once again refer to Nadathur[2].

3 Singular homology

Having gone through defining simplicial homology groups the process for singular homology is analogous, with the only difference being what the group of k -cycles looks like and how that connects to the boundary operator. Since we can not in general find a simplicial complex structure on arbitrary topological spaces there is no given set of simplices for us to use as a basis for C_k . Instead the basis of C_k for a space X will now be the set of *all* continuous functions from the k -simplex into X :

$$C_k = F(\{\sigma : \Delta^k \rightarrow X \mid \sigma \text{ continuous}\})$$

This basis should be thought of as all the ways we can put a k -simplex into X , including not only embeddings but also all those ways which have some kind of "singularity" (e.g. some non-injectivity, or more concretely mapping a simplex to a point). Another thing worth noting here is that we now talk about *the* k -simplex Δ^k , in the sense of the (up to homeomorphism) unique topological space that we get from any geometric realization of such a simplex. Furthermore all these functions σ that map Δ^k into X we will now refer to as X 's singular k -simplices.

The boundary operator ∂_k applied to a basis-element σ will once again be the formal sum of the boundaries of σ (i.e. the restrictions of σ to each of the faces of Δ^k), taking into account the orientation of the simplices:

$$\partial_k : C_k \rightarrow C_{k-1}, \quad \sigma \mapsto \sum_{i=0}^k (-1)^i \sigma|_{\Delta_i^k}$$

Where Δ_i^k denotes the face of Δ^k without its i th vertex and σ is a singular simplex of X . Since any face of Δ^k is a $(k-1)$ -simplex, the restriction to one such face $\sigma|_{\Delta_i^k}$ will indeed be another singular simplex, making it a valid element of C_{k-1} . Being defined on all the basis elements of C_k , ∂_k once again naturally extends to a group homomorphism.

The remainder of the construction of the singular homology groups is identical to the simplicial ones. We define them to be the quotients of the kernels of ∂_k (the k -cycles) and the images of ∂_{k+1} (the k -boundaries):

$$\begin{aligned} Z_k &= \ker(\partial_k) \\ B_k &= \text{im}(\partial_{k+1}) \\ H_k(X) &= Z_k / B_k \end{aligned}$$

References

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